UNIT 4 · 8 hours

Stability, Characteristic Equation and Root Location



Stability of Systems

Control Systems (22EEG53)
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Learning objectives

- Define stability of an LTI system
- Relate stability to the roots of the characteristic equation
- Distinguish absolute and relative stability
- Classify systems from root locations

Syllabus context

Stability of systems: characteristic equation, location of roots, absolute and relative stability, Routh–Hurwitz criterion (necessary and sufficient conditions), root locus concept, rules for construction of root locus.

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Definition (BIBO): a system is stable if every bounded input gives a bounded output.

Equivalent: the natural (impulse) response decays to zero as $t \to \infty$.

Unstable: the response grows without bound (e.g. an inverted pendulum without control).

Marginally stable: response neither decays nor grows: sustained oscillation.

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Characteristic equation

$$1 + G(s)H(s) = 0$$

characteristic equation

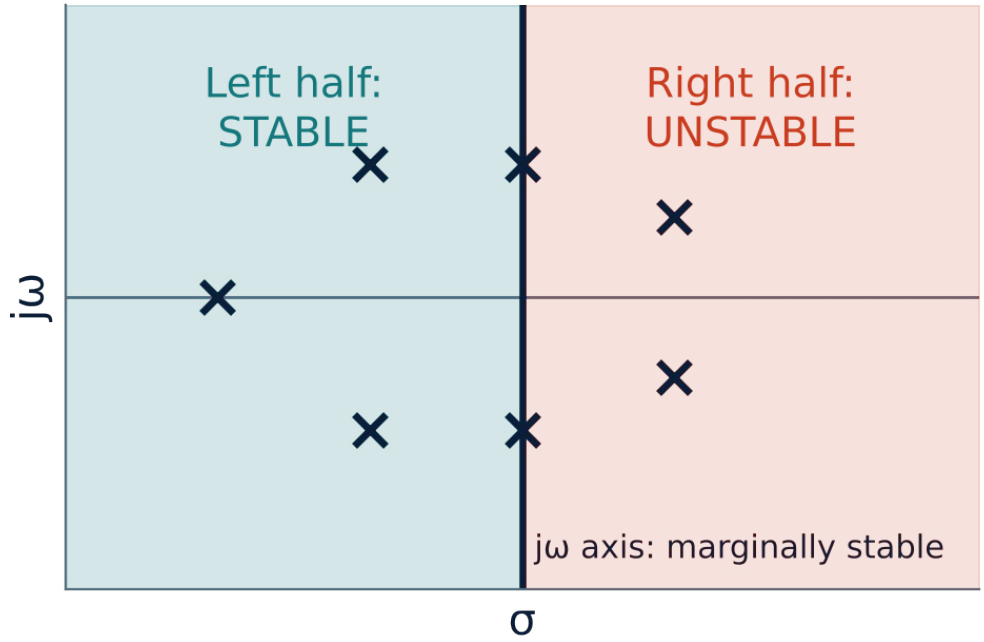
$$\Delta(s) = a0\, s^{n} + a1\, s^{n-1} + \ldots + an = 0$$

polynomial form

- Roots are the closed-loop poles; each pole gives a term in the response: e^(pt).
- A pole p = σ + jω gives e^(σt) × (oscillation): decays if σ < 0.

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Root location and stability



- All roots in the left half-plane (LHP): stable.
- Any root in the right half-plane (RHP): unstable.
- Simple roots on the jω axis (and none in the RHP): marginally stable.
- Repeated jω-axis roots: unstable.

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Examples of root locations

Characteristic equation	Roots	Stability
$s^2 + 3s + 2$	−1, −2	Stable
$s^2 + 4$	± j2	Marginally stable
$s^2 - s + 2$	0.5 ± j1.32	Unstable
$s^2 + 2s + 5$	−1 ± j2	Stable (underdamped)
$s^3 + 2s^2 + s + 2$	−2, ± j1	Marginally stable

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Absolute versus relative stability

Absolute stability

- Is the system stable or not?
- Decided by sign of the real parts of roots
- Tests: Routh–Hurwitz, root locus, Bode, Nyquist
- Binary answer: stable / unstable

Relative stability

- How stable? How close to the boundary?
- Measured by distance of roots from the jω axis, damping ratio, GM and PM
- Shifted-axis Routh test: $s = z - \sigma$
- Determines transient quality

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Necessary conditions

- For all roots to be in the LHP, all coefficients must be non-zero and have the same sign (necessary, not sufficient for n ≥ 3).
- Missing term or sign change ⇒ certainly not stable.
- For n = 1, 2 the condition is also sufficient.
- For higher order: Routh–Hurwitz criterion (next topic).

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Key takeaways

- Stable $\Leftrightarrow$ all closed-loop poles in the LHP
- Characteristic equation: $1 + GH = 0$
- $j\omega$-axis simple roots: marginal; RHP roots: unstable
- Absolute stability: yes/no; relative stability: how far from the boundary

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Practice questions

Q1. Define BIBO stability and relate it to pole locations.

Q2. Classify $s^3 + 4s^2 + 5s + 2$ using the coefficient test.

Q3. Distinguish absolute and relative stability.

Q4. Find the roots and the stability of $s^2 + 6s + 8$.